

Complex coordinates forced by finite orthogonality hypergraphs in dimension three

Mirko Navara¹  and Karl Svozil^{2,*} 

¹Faculty of Electrical Engineering, Czech Technical University in Prague,
Technická 2, CZ-166 27 Prague 6, Czech Republic

²Institute for Theoretical Physics, TU Wien,
Wiedner Hauptstraße 8–10/136, A-1040 Vienna, Austria

E-mail: navara@fel.cvut.cz; karl.svozil@tuwien.ac.at

*Author to whom correspondence should be addressed.

Abstract. We exhibit two finite orthogonality hypergraphs that admit faithful orthogonal representations by rays in complex three-dimensional space but admit none in real three-dimensional space. Here faithfulness requires distinct rays and excludes additional orthogonalities. Thus, although each orthogonality diagram is unchanged, its coordinatizability in dimension three depends on whether the scalar field is complex or real. The first hypergraph has 20 vertices and 12 contexts. The second hypergraph adds one vertex and two contexts. Phase-adjusted realification also gives explicit faithful orthogonal representations of both hypergraphs in real six-dimensional space. We discuss the physical meaning in terms of shared measurement outcomes, equivalent pure-state mixtures, and spin-1 polarization, while distinguishing the assumptions retained by larger real descriptions.

Keywords: quantum foundations, real quantum theory, orthogonality hypergraphs, qutrits, projective measurements

1 Introduction

Any single three-outcome projective measurement can be described with real vectors. The question is whether one common choice of coordinates can make an entire family of overlapping measurements real. The octagons studied here show that this can fail. Complex three-dimensional space accommodates the family, while real three-dimensional space does not. Physically, each vertex denotes a rank-one projector, and each context denotes a complete measurement. A repeated label must identify the same test, distinct labels must remain distinct, and both the stated orthogonalities and nonorthogonalities must be respected.

Relative phases between components therefore affect which measurement arrangements a three-level system can support. The first octagon also gives this distinction a preparation meaning: it forces two different equal mixtures of pure states to produce the same mixed

state. We return to this interpretation, its spin-1 realization, and the scope of a larger real description in Sec. 7.

An orthogonality hypergraph records which finite families of rays are required to form orthogonal bases. Its representability may depend on the scalar field, even when the ambient dimension is fixed. Harding and Salinas Schmeis constructed a finite orthogonality graph representable over $\mathbb{C}^3$ but not over $\mathbb{R}^3$ [1]. Counting the two 25-vertex, 16-context gadgets in their Fig. 2 and Theorem 2.6 gives 48 vertices and 32 specified triples when the edge joining their centers is completed to a triple. These are counts of their specified diagram; our faithfulness condition additionally excludes unlisted orthogonalities. The related Kochen–Specker (KS) completions analyzed by Navara and Svozil have 69 rays and 50 contexts [2]. The present 20-ray, 12-context and 21-ray, 14-context examples isolate the field separation through short projector-rigidity and closure arguments, without claiming minimum size. Conversion of KS sets into bipartite perfect quantum strategies is a separate problem studied by Trandafir and Cabello [3].

The role of complex scalars in quantum theory involves two questions that should not be conflated. If a change of dimension and an additional complex structure are allowed, a complex Hilbert space $\mathbb{C}^n$ can be represented on the real space $\mathbb{R}^{2n}$ by separating real and imaginary parts; related constructions reproduce broad classes of complex quantum protocols within a real formalism [4–6]. Network separations instead impose composition and source-independence assumptions, rather than a fixed Hilbert-space dimension [7, 8]. Other operational comparisons use specified discrimination tasks [9]. Here the comparison concerns single-system projective geometry at fixed dimension; no network or tensor-product assumption enters the proofs.

A phase-adjusted realification method [10] makes the boundary precise for finite orthogonality data. It preserves and reflects all pairwise orthogonalities in $\mathbb{R}^6$, but a three-ray context then spans only half of the ambient real space. Below we apply this construction explicitly to both octagons after proving their same-dimension separation.

We present two further exact finite examples, both with an octagonal incidence skeleton. The 20-vertex hypergraph H_{20} forces an equality of two sums of rank-one projectors. Complex phases permit two distinct nonorthogonal two-ray decompositions of the same positive operator; over the reals, the spectral decomposition fixes the rays. The 21-vertex hypergraph H_{21} is obtained by completing both inner pairs with a common ray. The spectral argument then becomes degenerate, but the perimeter closure relations supply a second obstruction to real coordinates.

Unlike a conventional Kochen–Specker set, neither hypergraph is used through an uncolorability argument. Its role here is instead to enforce exact orthogonality incidences and the resulting projector or closure identities; no contradiction involving two-valued states is assumed or needed.

The broader setting is quantum contextuality and its experimental and information-processing uses, reviewed by Budroni et al. [11]. Cabello’s 33-ray, 14-basis KS set [12] links a compact qutrit construction to bipartite perfect strategies. Our 12- and 14-context hypergraphs address a different question: both are colorable, yet their faithful rank-one measurement geometry separates $\mathbb{C}^3$ from $\mathbb{R}^3$. Their sizes therefore do not improve a KS minimality bound. Muller and Giorgetti [13] show how a hypergraph together with an anticommutation graph determines the contextuality degree of its admissible Pauli-

observable labelings. Here the vertices instead denote rays, and the constraints concern orthogonality and completeness of three-outcome measurements. The common theme is that finite incidence data constrain quantum realizations; the physical question here is which scalar field can support those realizations at fixed dimension.

The results established below go beyond merely exhibiting convenient complex coordinates. For each displayed representation we verify all unordered ray pairs exactly: the prescribed pairs are orthogonal, every other inner product is nonzero, and distinct vertices label distinct rays. Thus the word “faithful” carries its full graph-theoretic meaning [14, 15].

2 Orthogonality representations and two algebraic lemmas

Throughout, we use the physics convention for the Hermitian inner product, which is conjugate-linear in the first argument:

$$\langle x, y \rangle = \sum_{k=1}^d \overline{x_k} y_k. \quad (1)$$

Definition 1. Let $H = (V, \mathcal{E})$ be a three-uniform hypergraph, let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$, and let $d \geq 3$. Hyperedges are unordered three-element subsets of V , and no hyperedge is repeated. A *faithful orthogonal representation* of H in $\mathbb{F}^d$ assigns a nonzero vector $x_v \in \mathbb{F}^d$ to every $v \in V$ such that:

1. distinct vertices determine distinct rays; and
2. $\langle x_u, x_v \rangle = 0$ if and only if u and v belong to a common hyperedge.

Each hyperedge is then an orthogonal triple. For $d = 3$ it is an orthogonal basis and hence a maximal *context*; for $d > 3$ the same specified triple need not be maximal.

For a nonzero column vector x , let $x^\dagger$ denote its conjugate transpose and write

$$P_x = \frac{xx^\dagger}{\langle x, x \rangle} \quad (2)$$

for the rank-one projector onto its ray. In a three-dimensional representation, a context $\{x, y, z\}$ satisfies

$$P_x + P_y + P_z = I_3. \quad (3)$$

Consequently, integer linear combinations of three-dimensional context equations can force projector identities that hold in every representation of the hypergraph in that dimension.

For $x, y \in \mathbb{C}^3$ define the Hermitian cross product by

$$x \boxtimes y := \overline{x \times y}, \quad (4)$$

where $x \times y$ is the ordinary bilinear cross product. Then $x \boxtimes y$ is Hermitian-orthogonal to both x and y .

Lemma 2 (Cyclic Gram closure). *Let $x, y, z \in \mathbb{C}^3$, and suppose $x \boxtimes y$ and $y \boxtimes z$ are nonzero. Then*

$$\langle x \boxtimes y, y \boxtimes z \rangle = 0 \iff \langle x, y \rangle \langle y, z \rangle = \langle x, z \rangle \langle y, y \rangle. \quad (5)$$

Proof. The bilinear Lagrange identity, with $\cdot$ denoting the standard bilinear dot product on $\mathbb{C}^3$, gives

$$(x \times y) \cdot (\bar{y} \times \bar{z}) = (x \cdot \bar{y})(y \cdot \bar{z}) - (x \cdot \bar{z})(y \cdot \bar{y}).$$

Indeed, for arbitrary $a, b \in \mathbb{C}^3$, $\langle \bar{a}, \bar{b} \rangle = a \cdot \bar{b}$; taking $a = x \times y$ and $b = y \times z$ therefore identifies the left-hand side with $\langle x \boxtimes y, y \boxtimes z \rangle$. The right-hand side is the complex conjugate of

$$\langle x, y \rangle \langle y, z \rangle - \langle x, z \rangle \langle y, y \rangle.$$

Their vanishing is therefore equivalent. $\square$

Lemma 3 (Rigidity of two real rays). *Let a_1, a_2, b_1, b_2 be real unit vectors satisfying*

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}. \quad (6)$$

If a_1 and a_2 are distinct and nonorthogonal, then $\{[b_1], [b_2]\} = \{[a_1], [a_2]\}$ as unordered pairs of rays.

Proof. All four vectors lie in the range of the common operator in Eq. (6). Choose signs so that $\gamma = \langle a_1, a_2 \rangle \in (0, 1)$. The two nonzero eigenvalues are $1 + \gamma$ and $1 - \gamma$, with eigenvectors $a_1 + a_2$ and $a_1 - a_2$, respectively. Choose signs and order for the b -vectors so that $\delta = \langle b_1, b_2 \rangle \geq 0$. The common operator has rank two, so $\delta < 1$. Equality of spectra gives $\delta = \gamma$. Since the two eigenvalues are distinct, there are real nonzero scalars λ, μ such that

$$b_1 + b_2 = \lambda(a_1 + a_2), \quad b_1 - b_2 = \mu(a_1 - a_2).$$

The unit-norm and inner-product equations give

$$\begin{aligned} \lambda^2(1 + \gamma) + \mu^2(1 - \gamma) &= 2, \\ \lambda^2(1 + \gamma) - \mu^2(1 - \gamma) &= 2\gamma. \end{aligned}$$

Hence $\lambda^2 = \mu^2 = 1$. Solving for b_1, b_2 recovers the two original rays, possibly with signs and permutation. $\square$

3 The 20-ray octagon

The vertices of H_{20} are eight midpoint vertices $v_0, \dots, v_7$, eight corner vertices $v_{01}, v_{12}, \dots, v_{70}$, and four inner vertices a_1, a_2, b_1, b_2 . Indices are read modulo eight. The eight perimeter contexts and four inner contexts are

$$E_j = \{v_{j-1,j}, v_j, v_{j,j+1}\}, \quad j = 0, \dots, 7, \quad (7)$$

$$\begin{aligned} D_0 &= \{v_0, a_1, v_4\}, & D_2 &= \{v_2, a_2, v_6\}, \\ D_1 &= \{v_1, b_2, v_5\}, & D_3 &= \{v_3, b_1, v_7\}. \end{aligned} \quad (8)$$

The incidence structure is shown in Fig. 1. Table 1 lists all contexts of both octagons.

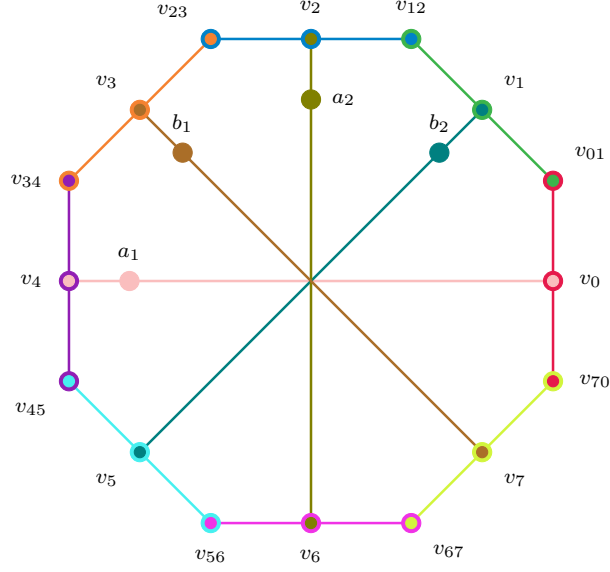


Figure 1. The 20-ray hypergraph H_{20} . The eight perimeter contexts $E_0, \dots, E_7$ form the octagon; the four inner contexts join opposite midpoints through a_1, a_2, b_1, b_2 . Each color denotes one context.

Table 1. The twelve contexts of H_{20} and the two additional contexts of H_{21} . All fourteen contexts belong to H_{21} ; only F_a and F_b are absent from H_{20} .

Context	Vertices (unordered)
E_0	$\{v_{70}, v_0, v_{01}\}$
E_1	$\{v_{01}, v_1, v_{12}\}$
E_2	$\{v_{12}, v_2, v_{23}\}$
E_3	$\{v_{23}, v_3, v_{34}\}$
E_4	$\{v_{34}, v_4, v_{45}\}$
E_5	$\{v_{45}, v_5, v_{56}\}$
E_6	$\{v_{56}, v_6, v_{67}\}$
E_7	$\{v_{67}, v_7, v_{70}\}$
D_0	$\{v_0, a_1, v_4\}$
D_1	$\{v_1, b_2, v_5\}$
D_2	$\{v_2, a_2, v_6\}$
D_3	$\{v_3, b_1, v_7\}$
F_a	$\{a_1, c, a_2\}$
F_b	$\{b_1, c, b_2\}$

3.1 The covering identity

Consider the two collections of six contexts

$$\mathcal{A} = \{D_0, D_2, E_1, E_3, E_5, E_7\},$$

$$\mathcal{B} = \{D_1, D_3, E_0, E_2, E_4, E_6\}.$$

Every midpoint and every corner occurs exactly once in each collection. Only the two a -vertices occur in $\mathcal{A}$, and only the two b -vertices occur in $\mathcal{B}$. Summing Eq. (3) over the two collections and subtracting therefore gives

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}. \quad (9)$$

This identity is combinatorial: it holds in every real or complex three-dimensional representation of H_{20} .

3.2 An exact faithful representation over $\mathbb{C}^3$

Set

$$\begin{aligned} C &= \frac{49 - 5\sqrt{69}}{26}, \quad s = \sqrt{C}, \\ N &= 1 + C, \quad r = \sqrt{N}. \end{aligned} \quad (10)$$

The number C is the smaller positive root of

$$13C^2 - 49C + 13 = 0, \quad (11)$$

so $0 < C < 1$. Define the unit phases

$$\eta = \frac{5 + 12i}{13}, \quad (12)$$

$$\zeta = \frac{14 + 6\sqrt{69}}{65} + i\frac{4\sqrt{69} - 21}{65}. \quad (13)$$

Let

$$\begin{aligned} (\mu_0, \dots, \mu_7) &= (-1, \eta, 1, -\eta, -1, \eta, 1, -\eta), \\ (\tau_0, \dots, \tau_7) &= (1, \zeta, -i, -i\zeta, -1, -\zeta, i, i\zeta). \end{aligned}$$

The inner and midpoint rays are represented by

$$\begin{aligned} a_1 &= (s, 1, 0), & a_2 &= (s, -1, 0), \\ b_1 &= (s, \eta, 0), & b_2 &= (s, -\eta, 0), \end{aligned} \quad (14)$$

$$v_j = (1, s\mu_j, r\tau_j), \quad j = 0, \dots, 7. \quad (15)$$

For j modulo eight, define

$$v_{j,j+1} = v_j \boxtimes v_{j+1}. \quad (16)$$

Equivalently, Eq. (16) is the explicit formula

$$\begin{aligned} v_{j,j+1} &= (sr(\bar{\mu}_j\bar{\tau}_{j+1} - \bar{\tau}_j\bar{\mu}_{j+1}), \\ &\quad r(\bar{\tau}_j - \bar{\tau}_{j+1}), s(\bar{\mu}_{j+1} - \bar{\mu}_j)). \end{aligned} \quad (17)$$

Theorem 4. *Equations (10)–(17) give a faithful orthogonal representation of H_{20} in $\mathbb{C}^3$.*

Proof. First, $|\eta| = 1$. Write $\zeta = x + iy$. The four inner contexts follow directly from $r^2 = N$, $|\eta| = |\zeta| = 1$, and the antipodal relations $\mu_{j+4} = \mu_j$, $\tau_{j+4} = -\tau_j$. For example,

$$\langle a_1, v_0 \rangle = 0, \quad \langle v_0, v_4 \rangle = 1 + C - r^2 = 0,$$

and

$$\langle b_2, v_1 \rangle = 0, \quad \langle v_1, v_5 \rangle = 1 + C - r^2 |\zeta|^2 = 0.$$

The other two inner contexts are identical calculations.

By construction, $v_{j,j+1}$ is orthogonal to v_j and v_{j+1} . It remains to prove orthogonality of the two corner rays in every perimeter context. Put

$$G_{jk} = \langle v_j, v_k \rangle = 1 + C \bar{\mu}_j \mu_k + N \bar{\tau}_j \tau_k.$$

Lemma 2 reduces the eight remaining conditions to

$$G_{j-1,j} G_{j,j+1} = G_{j-1,j+1} G_{j,j}, \quad j = 0, \dots, 7. \quad (18)$$

Substituting the cyclic patterns for μ_j and τ_j reduces all eight conditions to common real and imaginary equations once $x^2 + y^2 = 1$. Separating those parts gives

$$7Cx - 17Cy - 13C - 13x + 13y + 13 = 0, \quad (19)$$

and

$$\begin{aligned} -7C^2x + 17C^2y + 13C^2 - 20Cx + 30Cy \\ + 2C - 13x + 13y + 13 = 0. \end{aligned} \quad (20)$$

Solving these two linear equations for x and y gives

$$x = \frac{104 - 76C}{65(1 + C)}, \quad y = \frac{39 - 81C}{65(1 + C)}. \quad (21)$$

For these values,

$$x^2 + y^2 - 1 = \frac{48(13C^2 - 49C + 13)}{325(1 + C)^2} = 0. \quad (22)$$

Using the smaller root in Eq. (11), Eq. (21) is exactly the phase in Eq. (13). Thus all twelve triples are contexts.

It remains to exclude unintended orthogonalities and ray coincidences. The exact overlap certificate in Appendix A shows that the 36 prescribed unordered pairs have zero overlap, whereas the 154 unprescribed pairs have squared normalized overlaps between $1/25$ and $4/5$. Hence every unprescribed inner product is nonzero and no two displayed vectors are proportional. The representation is faithful. $\square$

The two inner pairs indeed decompose the same operator. Since all four vectors have squared norm N and $|\eta| = 1$,

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2} = \frac{2}{N} \text{diag}(C, 1, 0). \quad (23)$$

Their within-pair inner products are $C - 1 \neq 0$, and the four rays are distinct because $\eta \notin \{1, -1\}$.

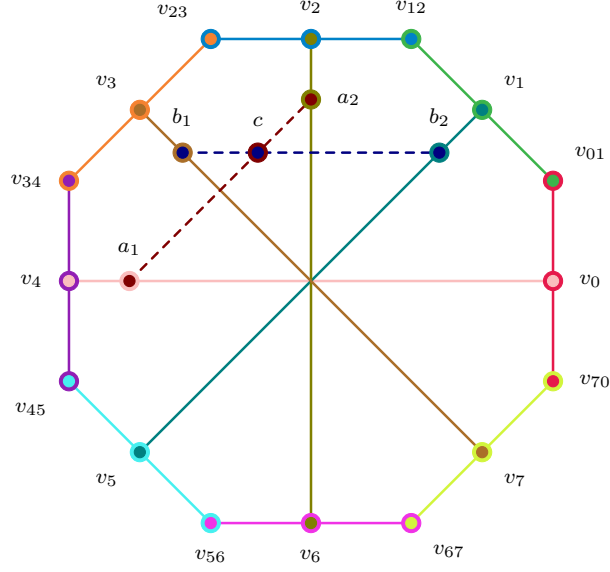


Figure 2. The 21-ray hypergraph H_{21} . The twelve solid contexts are those of H_{20} ; the two dashed contexts meet at the new vertex c .

3.3 Nonexistence over $\mathbb{R}^3$

Theorem 5. *The hypergraph H_{20} has no faithful orthogonal representation in $\mathbb{R}^3$.*

Proof. Suppose a faithful real representation existed. The covering argument is independent of the field, so Eq. (9) would hold. Faithfulness implies that a_1, a_2 are distinct and nonorthogonal, because they do not share a context; the same is true of b_1, b_2 . Lemma 3 then identifies the unordered pair of b -rays with the unordered pair of a -rays. This contradicts the requirement that four distinct vertices label four distinct rays. $\square$

4 The 21-ray orthogonal augmentation

Add one vertex c and two contexts

$$F_a = \{a_1, c, a_2\}, \quad F_b = \{b_1, c, b_2\} \quad (24)$$

to H_{20} . The result is the 21-vertex, 14-context hypergraph H_{21} shown in Fig. 2. The added contexts force both inner pairs to be orthogonal and to have the same one-dimensional orthogonal complement.

4.1 Exact coordinates over $\mathbb{C}^3$

Let

$$p = \frac{\sqrt{14} + \sqrt{2}}{4}, \quad q = \frac{\sqrt{14} - \sqrt{2}}{4}. \quad (25)$$

Then

$$p^2 + q^2 = 2, \quad pq = \frac{3}{4}, \quad p - q = \frac{\sqrt{2}}{2}. \quad (26)$$

Choose

$$\begin{aligned}
c &= (0, 0, 1), \\
a_1 &= (1, 1, 0), & a_2 &= (1, -1, 0), \\
b_1 &= (1, i, 0), & b_2 &= (1, -i, 0),
\end{aligned} \tag{27}$$

and the eight midpoint representatives

$$\begin{aligned}
v_0 &= (1, -1, \sqrt{2}), & v_4 &= (1, -1, -\sqrt{2}), \\
v_1 &= (1, i, p + iq), & v_5 &= (1, i, -p - iq), \\
v_2 &= (1, 1, -i\sqrt{2}), & v_6 &= (1, 1, i\sqrt{2}), \\
v_3 &= (1, -i, q - ip), & v_7 &= (1, -i, -q + ip).
\end{aligned} \tag{28}$$

The eight corner representatives are

$$\begin{aligned}
v_{01} &= (-p + i(q + \sqrt{2}), \sqrt{2} - p + iq, 1 - i), \\
v_{12} &= (\sqrt{2} - p + iq, p - i(q + \sqrt{2}), 1 + i), \\
v_{23} &= (q + \sqrt{2} + ip, -q + i(\sqrt{2} - p), -1 + i), \\
v_{34} &= (q + i(p - \sqrt{2}), q + \sqrt{2} + ip, -1 - i), \\
v_{45} &= (p - i(q + \sqrt{2}), -\sqrt{2} + p - iq, 1 - i), \\
v_{56} &= (-\sqrt{2} + p - iq, -p + i(q + \sqrt{2}), 1 + i), \\
v_{67} &= (-q - \sqrt{2} - ip, q - i(\sqrt{2} - p), -1 + i), \\
v_{70} &= (-q - i(p - \sqrt{2}), -q - \sqrt{2} - ip, -1 - i).
\end{aligned} \tag{29}$$

Theorem 6. *Equations (25)–(29) give a faithful orthogonal representation of H_{21} in $\mathbb{C}^3$.*

Proof. The two added contexts are immediate from Eq. (27). For the original inner contexts, representative calculations are

$$\begin{aligned}
\langle a_1, v_0 \rangle &= 0, & \langle v_0, v_4 \rangle &= 1 + 1 - 2 = 0, \\
\langle b_2, v_1 \rangle &= 0, & \langle v_1, v_5 \rangle &= 2 - (p^2 + q^2) = 0.
\end{aligned}$$

The two remaining inner contexts follow in the same way.

Direct evaluation of the Hermitian cross product gives, exactly,

$$v_{j,j+1} = v_j \boxtimes v_{j+1}$$

for each representative in Eq. (29). Hence every corner is orthogonal to its two incident midpoints. Substitution in the Gram closure identity (5) reduces the eight remaining perimeter inner products to the identities in Eq. (26); they therefore vanish.

Finally, Appendix A gives an exact check of all 210 unordered pairs. The 42 pairs prescribed by the fourteen contexts have zero overlap. Every remaining pair has squared normalized overlap either 1/4 or 1/2. Thus there are no additional orthogonalities and no ray coincidences. $\square$

4.2 Nonexistence over $\mathbb{R}^3$

Theorem 7. *The hypergraph H_{21} has no faithful orthogonal representation in $\mathbb{R}^3$.*

Proof. Assume that a faithful real representation exists. A global orthogonal transformation and independent rescalings of ray representatives allow us to set

$$c = (0, 0, 1), \quad a_1 = (1, 0, 0), \quad a_2 = (0, 1, 0). \quad (30)$$

The second added context lies in the plane $c^\perp$, so for some angle θ ,

$$b_1 = (\cos \theta, \sin \theta, 0), \quad b_2 = (-\sin \theta, \cos \theta, 0). \quad (31)$$

Faithfulness excludes $\theta \in \frac{\pi}{2}\mathbb{Z}$, since those values identify a b -ray with an a -ray. Put

$$m = \tan \theta \neq 0, \quad M = 1 + m^2. \quad (32)$$

Write a representative vector of a midpoint ray as $v_j = (X_j, Y_j, Z_j)$. No midpoint ray can lie in $c^\perp$. Indeed, a planar vector orthogonal to the inner ray in its context is proportional to the other member of one of the two planar bases $\{a_1, a_2\}$ or $\{b_1, b_2\}$, contradicting faithfulness. Thus $Z_j \neq 0$ for every j , and each representative may be rescaled to

$$v_j = (X_j, Y_j, 1). \quad (33)$$

The inner contexts D_0, D_1, D_3 imply

$$X_0 = X_4 = 0, \quad Y_0 Y_4 + 1 = 0, \quad (34)$$

$$Y_1 = mX_1, \quad Y_5 = mX_5, \quad MX_1 X_5 + 1 = 0, \quad (35)$$

$$X_3 = -mY_3, \quad X_7 = -mY_7, \quad MY_3 Y_7 + 1 = 0. \quad (36)$$

In particular,

$$X_5 = -\frac{1}{MX_1}, \quad Y_7 = -\frac{1}{MY_3}, \quad (37)$$

and $X_1 Y_3 \neq 0$.

The corner rays can be eliminated from the perimeter by the real form of Lemma 2. At $j = 0$, Eq. (5) factors as

$$Y_0((mX_1 Y_7 - 1)Y_0 + Y_7 + mX_1) = 0. \quad (38)$$

The first factor cannot vanish because of Eq. (34). Moreover, the coefficient $1 - mX_1 Y_7$ cannot vanish: otherwise Eq. (38) would also give $Y_7 + mX_1 = 0$, and the two relations would imply $-m^2 X_1^2 = 1$. Therefore

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1 Y_7}. \quad (39)$$

The closure equation at $j = 4$ gives analogously

$$Y_4 = \frac{Y_3 + mX_5}{1 - mY_3 X_5}, \quad (40)$$

with a nonzero denominator.

Substituting Eq. (37) into Eqs. (39) and (40) yields

$$Y_0 = \frac{mMX_1Y_3 - 1}{MY_3 + mX_1}, \quad Y_4 = \frac{MX_1Y_3 - m}{MX_1 + mY_3}. \quad (41)$$

Set $u = X_1Y_3$. The remaining equation $Y_0Y_4 = -1$ becomes

$$(mMu - 1)(Mu - m) = -(mX_1 + MY_3)(MX_1 + mY_3). \quad (42)$$

Expanding the left-hand side gives

$$\begin{aligned} (mMu - 1)(Mu - m) &= mM^2u^2 - M(m^2 + 1)u + m \\ &= mM^2u^2 - M^2u + m, \end{aligned}$$

where the second equality uses $M = 1 + m^2$. The right-hand side is

$$-u(M^2 + m^2) - mM(X_1^2 + Y_3^2).$$

After cancellation and division by $m \neq 0$, Eq. (42) is therefore equivalent to the master equation

$$M^2u^2 + mu + 1 + M(X_1^2 + Y_3^2) = 0. \quad (43)$$

Since X_1, Y_3 are real, $X_1^2 + Y_3^2 \geq 2|X_1Y_3| = 2|u|$. The left-hand side of Eq. (43) is consequently at least

$$\begin{aligned} M^2u^2 + mu + 1 + 2M|u| &\geq 1 + M^2u^2 + |u|(2M - |m|) \\ &\geq 1. \end{aligned}$$

Here, with $t = |m|$,

$$2M - |m| = 2t^2 - t + 2 = 2\left(t - \frac{1}{4}\right)^2 + \frac{15}{8} > 0.$$

This contradicts Eq. (43), and the assumed real representation cannot exist. $\square$

5 Faithful realification in six dimensions

Theorems 5 and 7 keep the ambient dimension fixed. When dimension doubling is allowed, both hypergraphs have faithful real representations. We first recall the finite construction and then give explicit phase choices for the present coordinates [10].

Theorem 8 (Phase-adjusted realification). *Every finite faithful orthogonal representation by rays in $\mathbb{C}^3$ induces a faithful orthogonal representation of the same hypergraph by rays in $\mathbb{R}^6$.*

Proof. Let $\psi_1, \dots, \psi_N \in \mathbb{C}^3$ be nonzero representatives and put $c_{k\ell} = \langle \psi_k, \psi_\ell \rangle$. Define the canonical realification

$$\Phi_0(z_1, z_2, z_3) = (\Re z_1, \Re z_2, \Re z_3, \Im z_1, \Im z_2, \Im z_3). \quad (44)$$

For phases $\theta_k \in \mathbb{R}$, set $R_k = \Phi_0(e^{i\theta_k}\psi_k)$. Direct expansion gives

$$R_k \cdot R_\ell = \Re(e^{i(\theta_\ell - \theta_k)} c_{k\ell}). \quad (45)$$

Thus $c_{k\ell} = 0$ implies $R_k \cdot R_\ell = 0$ for every phase choice. If $c_{k\ell} \neq 0$, write $c_{k\ell} = |c_{k\ell}|e^{i\varphi_{k\ell}}$. The right-hand side of Eq. (45) vanishes precisely when

$$\theta_\ell - \theta_k + \varphi_{k\ell} \equiv \frac{\pi}{2} \pmod{\pi}. \quad (46)$$

Choose $\theta_1, \dots, \theta_N$ inductively. Suppose that the first $m-1$ phases have been fixed. For each $k < m$ with $c_{km} \neq 0$, the unwanted orthogonality equation $R_k \cdot R_m = 0$ excludes, by Eq. (46), exactly two values of θ_m modulo 2π . Their union is finite, so a phase outside it exists. Continuing through $m = N$ makes the real dot product nonzero for every originally nonorthogonal pair, while all original orthogonalities remain zero.

It remains only to check ray distinctness. If R_k and R_ℓ were real proportional, then $e^{i\theta_k}\psi_k$ and $e^{i\theta_\ell}\psi_\ell$ would be real proportional and hence would represent the same complex ray. Faithfulness of the original representation excludes this. Therefore the realified representation is faithful. $\square$

For explicit coordinates it is convenient to parametrize the phase by a real number t :

$$\lambda_t = \frac{1+it}{\sqrt{1+t^2}}, \quad \Phi_t(z) = \Phi_0(\lambda_t z). \quad (47)$$

Writing $z = x + iy$ with $x, y \in \mathbb{R}^3$ gives the entirely real formula

$$\Phi_t(x + iy) = \frac{(x - ty, y + tx)}{\sqrt{1+t^2}} \in \mathbb{R}^6. \quad (48)$$

Moreover,

$$\Phi_{t_u}(u) \cdot \Phi_{t_v}(v) = \Re \left(\frac{(1 - it_u)(1 + it_v)}{\sqrt{(1+t_u^2)(1+t_v^2)}} \langle u, v \rangle \right). \quad (49)$$

Corollary 9 (Explicit real six-dimensional coordinates). *Let $\psi_v^{(20)}$ denote the representatives in Eqs. (10)–(17). Define*

$$t_v^{(20)} = \begin{cases} 1, & v \in \{v_{34}, v_{45}, v_{56}\}, \\ 2, & v \in \{v_{67}, v_{70}\}, \\ 0, & \text{otherwise.} \end{cases} \quad (50)$$

Then $R_v^{(20)} = \Phi_{t_v^{(20)}}(\psi_v^{(20)})$ is a faithful orthogonal representation of H_{20} in $\mathbb{R}^6$.

Likewise, let $\psi_v^{(21)}$ denote the representatives in Eqs. (25)–(29), and define

$$t_v^{(21)} = \begin{cases} 1, & v = v_{45}, \\ 2, & v \in \{v_2, v_3, v_6, v_7, v_{34}, v_{56}, v_{70}\}, \\ 3, & v = v_{67}, \\ 0, & \text{otherwise.} \end{cases} \quad (51)$$

Then $R_v^{(21)} = \Phi_{t_v^{(21)}}(\psi_v^{(21)})$ is a faithful orthogonal representation of H_{21} in $\mathbb{R}^6$.

Proof. Equations (48), (50), and (51) give all six real coordinates directly from the complex vectors already displayed. Exact substitution in Eq. (49) gives zero for exactly 36 of the $\binom{20}{2} = 190$ pairs in H_{20} , namely the pairs prescribed by its twelve contexts. The other 154 real dot products are nonzero. For H_{21} , exactly 42 of the $\binom{21}{2} = 210$ real dot products vanish, namely the pairs prescribed by its fourteen contexts, and the other 168 are nonzero. These calculations take place in the algebraic fields generated by the radicals in the complex coordinates and by $\sqrt{1+t^2}$; no numerical tolerance is involved. Ray distinctness follows from Theorem 8. $\square$

Each specified hyperedge is therefore an orthogonal triple in $\mathbb{R}^6$, but it spans only a three-dimensional subspace. Its projector sum is the rank-three projector onto that subspace, not I_6 . This loss of maximality is precisely why the real six-dimensional representations do not contradict the two real three-dimensional no-go theorems.

6 Discussion

The two nonexistence proofs expose different real-complex mechanisms. For H_{20} , the context covering forces a rank-two positive operator to have two disjoint, nonorthogonal decompositions into two rank-one projectors. Complex relative phases make this possible. Over the reals, the two simple eigenvalues determine the two rays, producing the rigidity of Lemma 3.

For H_{21} , both inner pairs are orthogonal and their common projector sum is the identity on $c^\perp$. The eigenvalue argument is therefore unavailable: the relevant eigenvalue is degenerate and many real orthogonal bases of $c^\perp$ exist. The obstruction instead comes from compatibility with the octagonal perimeter. The two reduced closure equations determine Y_0 and Y_4 , while the inner context D_0 demands $Y_0 Y_4 = -1$; together they produce the positive master expression in Eq. (43).

Both statements are dimension-specific. Corollary 9 realizes the same pairwise orthogonality relations faithfully in $\mathbb{R}^6$, but the specified triples are no longer complete bases. Consequently, the three-dimensional context equation (3) is replaced by a rank-three subspace projector. The complementary covering no longer forces Eq. (9), and the three-dimensional cross-product closure used for H_{21} no longer applies. The comparison therefore concerns the scalar field at fixed dimension and fixed context maximality, not the possibility of simulating complex coordinates in a larger real space. It is also independent of KS noncolorability: the contradictions use orthogonality, completeness of contexts, and field-dependent algebra, not an impossibility of assigning two-valued states.

For H_{20} , exhaustive enumeration yields exactly 26 two-valued states, assigning values in $\{0, 1\}$ to the vertices with exactly one value 1 in each context. This family is unital, separating, and order determining. Thus the failure of faithful representability in $\mathbb{R}^3$ coexists with strong classical state-separation properties: the obstruction concerns the scalar field at fixed dimension, rather than the existence or separating power of two-valued states. These properties alone do not imply that every quantum probability assignment belongs to the convex hull of the 26 two-valued states, and hence do not by themselves exclude violations of noncontextuality inequalities.

For each hypergraph, the least dimension of a faithful real ray representation satisfies $4 \leq d_{\mathbb{R}} \leq 6$. We do not determine whether $\mathbb{R}^4$ or $\mathbb{R}^5$ suffices. Thus dimension doubling is a sufficient construction, not an established minimum. Nor do we establish the fewest vertices or contexts needed for this kind of separation. Although the H_{21} contradiction uses only part of its incidences, deleting contexts from the displayed realization can leave additional orthogonalities and destroy faithfulness.

Faithfulness is essential. Both hypergraphs admit a three-coloring in which each specified triple has all three colors; a checked coloring is included with the verification code. Assigning one real Cartesian basis ray to each color gives a nonfaithful representation in $\mathbb{R}^3$. It respects every specified orthogonality but identifies distinct vertices and adds other orthogonalities. The no-go theorems exclude faithful representations, not these collapsed ones.

7 Physical meaning

The covering identity of H_{20} has a direct preparation meaning. Suppose we toss a fair coin and prepare either a_1 or a_2 . We can instead prepare either b_1 or b_2 , again with equal probabilities. Once the preparation record is forgotten, both procedures produce the same density operator, namely one half of the common projector sum in Eq. (9). No subsequent measurement on the system can distinguish them. Yet the two pairs contain different pure states. Complex amplitudes allow this freedom. For real pure states, a mixture of two distinct, nonorthogonal states determines its equal-weight pair uniquely. The surrounding measurements force precisely the equality that exposes this difference.

The geometry is already visible in their common two-dimensional support. On the Bloch sphere, a fair mixture is the midpoint of the chord joining its two pure states. Away from the center, the sphere admits a circle's worth of chords with that midpoint. In a fixed real-state disk, exactly one chord has it as midpoint. Equal weights and a noncentral midpoint are essential: unequal weights, or the maximally mixed state at the center, remove this uniqueness.

The pairs are *pseudocontexts*: their summed test probabilities agree for every preparation, although neither pair consists of mutually exclusive outcomes of a single measurement [16]. These probabilities refer to separate tests. Their sum can exceed one and is not the probability of a single coarse-grained outcome. Its equality follows from how these tests participate in the surrounding complete measurements. Pseudocontexts thus connect different collections of tests through an equality of their aggregate statistics and, through equal mixing, connect different preparation procedures.

The construction H_{21} shows that the need for complex amplitudes extends beyond this particular preparation freedom. Its inner pairs are orthogonal, and a real plane readily accommodates different orthogonal bases. The difficulty appears when these bases must also fit the measurements around the perimeter. What fails is the consistency of the whole arrangement. Complex relative phases supply possibilities for fitting the measurements together that real vectors cannot supply.

A concrete interpretation is obtained with general projective measurements on a physical spin-1 system. In the Cartesian representation, a normalized pure state $\psi = x + iy$, with

real x, y , has spin expectation $\langle \mathbf{L} \rangle = 2\hbar x \times y$ [17]. This vanishes precisely when the ray has a real representative. Every faithful spin-1 realization of either octagon must therefore contain an outcome projector whose associated pure state has nonzero spin polarization. In the displayed H_{20} realization, the same mixed state can be prepared either from the two pure states a_1, a_2 with zero spin expectation or from the two pure states b_1, b_2 with equal and opposite nonzero spin expectations. In both preparations the mean spin vanishes.

A larger real description remains possible, but what it preserves matters. The six-dimensional construction in Sec. 5 uses real rays and preserves the orthogonality diagram; its three-ray contexts are then incomplete. A different, standard real construction preserves all probabilities and complete three-outcome measurements by replacing each complex ray with a two-dimensional real outcome subspace [5, 6]. The physical comparison therefore involves both the dimension and what counts as an elementary outcome.

The experimental prospect is a finite test of real three-dimensional models using preparations and sharp measurements. Such a test must establish the relevant dimension, the rank-one character of the outcomes, and the identity of tests shared between contexts. Distinct tests must also be distinguished by their statistics on suitable preparations, and the required nonorthogonalities must be retained within error bounds. The exact constructions identify a target for that comparison. Turning them into an experimental exclusion requires quantitative bounds on how closely a real model can reproduce the measured data.

8 Conclusion

The hypergraphs H_{20} and H_{21} give two exact finite separations between complex and real three-dimensional orthogonality representations. The first is governed by the field dependence of a two-projector decomposition; the second survives at the orthogonal limit and is governed by cyclic closure. Explicit algebraic vectors, complete nonexistence proofs, and exact pairwise-overlap certificates establish both results without numerical or coloring assumptions. Explicit phase-adjusted coordinates give faithful real representations in $\mathbb{R}^6$, where the specified triples are incomplete. This is an upper bound on the required real dimension.

The physical content is the need for complex relative phases in a common description of a finite measurement family. Pseudocontexts make this visible through equivalent preparation procedures; the spin-1 reading connects it to polarized outcome states. Larger real simulations clarify which assumptions about dimension and elementary outcomes carry the distinction.

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Conflict of interest

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Data availability statement

No experimental data were generated. The exact verification code and computed pairwise certificates are supplied with the manuscript.

A Exact faithfulness certificates

For two nonzero representatives x, y , define their squared normalized overlap by

$$Q(x, y) = \frac{|\langle x, y \rangle|^2}{\langle x, x \rangle \langle y, y \rangle}. \quad (\text{A1})$$

Thus $Q = 0$ exactly for orthogonal rays, while $Q = 1$ exactly for coincident rays. All calculations below are exact reductions in the algebraic fields generated by the displayed radicals and phases. The reductions and pair counts were independently cross-checked by exact symbolic arithmetic; no floating-point tolerance is used. The accompanying `verify_octagons.py` reproduces every pairwise overlap, both multiplicity tables, and the realification checks. It uses only the Python standard library, exact rational arithmetic, and radical relations; rational interval bounds certify the nonzero real expressions. The complete 400-pair output is supplied as `all-pairs.csv`.

For H_{20} , the 12 contexts prescribe 36 distinct unordered orthogonal pairs. Substitution of Eqs. (10)–(17), followed by reduction with Eq. (11), gives $Q = 0$ for precisely those 36 pairs. For the other 154 pairs, the complete list of values and multiplicities is:

Q	mult.	Q	mult.	Q	mult.
$\frac{1}{25}$	2	$\frac{3}{10}$	8	$\frac{8}{15}$	8
$\frac{3}{50}$	4	$\frac{23}{75}$	2	$\frac{14}{25}$	8
$\frac{7}{75}$	8	$\frac{49}{150}$	8	$\frac{16}{25}$	2
$\frac{8}{75}$	8	$\frac{26}{75}$	8	$\frac{7}{10}$	8
$\frac{7}{50}$	8	$\frac{28}{75}$	8	$\frac{59}{75}$	2
$\frac{4}{25}$	8	$\frac{32}{75}$	4	$\frac{4}{5}$	16
$\frac{1}{5}$	16	$\frac{7}{15}$	8		
$\frac{6}{25}$	8	$\frac{13}{25}$	2		

(A2)

The multiplicities sum to 154. In particular,

$$\frac{1}{25} \leq Q(x, y) \leq \frac{4}{5} \quad (\text{A3})$$

for every unprescribed pair. This proves simultaneously that no such pair is orthogonal and that no two vertices label the same ray.

For H_{21} , the 14 contexts prescribe 42 distinct unordered orthogonal pairs. Exact substitution of Eqs. (25)–(29) gives zero for precisely those pairs. The other 168 pairs split as

$$\frac{Q}{\text{multiplicity}} \left| \begin{array}{cc} \frac{1}{4} & \frac{1}{2} \\ 84 & 84 \end{array} \right. . \quad (\text{A4})$$

Again every unprescribed overlap lies strictly between zero and one. This completes the exact faithfulness verification used in Theorems 4 and 6.

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